

Computing the Sentinel-1 input to the decoder

Soil moisture project

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Scope. This document defines one thing only: what is computed from the Sentinel-1 archive and handed to each stage of the decoder. It does not discuss targets, losses, auxiliary heads, terrain or thermal data.

1 Why anything is needed

The decoder turns a 14×14 grid into a 112×112 map — 196 values per channel becoming 12,544. Every one of those values must come from somewhere.

In a conventional U-Net it comes from the skip connections: the encoder passed through 112×112 on the way down and saved a copy. TerraMind is a Vision Transformer. It cuts the image into 16×16 -pixel patches once, at the start, and every subsequent layer operates on those 196 patches. It never visits 112×112 , so there is no 112×112 copy to save — L3, L6, L9 and L12 are **all** 14×14 (model.py:530).

At model.py:253-255 those skips are FiLM-modulated; at model.py:257-264 they are interpolated up and concatenated. So every decoder stage receives 160 m information relabelled as 80/40/20 m. **Between 160 m and**

20 m the decoder has no measurement at all, and fills the gap from learned convolution weights. §23 measured what it fills it with: confident structure up to $0.19 \text{ m}^3/\text{m}^3$, strongest at the *flattest* site.

Sentinel-1 is the only dense observation available that varies in **both** space and time. It sits on disk as a 224×224 raster of σ^0 in dB per acquisition, median 440 acquisitions per station.

```
s1_asc 993 stations median 212 p10 90 max 958
s1_desc 910 stations median 238 p10 5 max 956
asc+desc median 440 p10 144 only 9 stations below 50

/projects/prjs1968/satellite_zarr/{station}.zarr/s1_{asc,desc}/
  data (N, 2, 224, 224) fp16, dB, bands [VV, VH]
  dates (N,) b'YYYYMMDD'
```

2 Notation

symbol	meaning
p	acquisition (pass) index, $p = 1 \dots n$
k	cell index at a given grid level
ℓ	decoder level, $\ell \in \{14, 28, 56, 112\}$
f_ℓ	pixels per cell, $f_\ell = 224/\ell$
B_k	the block of f_ℓ^2 pixels belonging to cell k
P	backscatter in linear power
T_ℓ	backscatter in dB after multilooking to level ℓ

The four levels correspond to the four decoder stages:

$$\ell = 14 \rightarrow 160 \text{ m}, \quad 28 \rightarrow 80 \text{ m}, \quad 56 \rightarrow 40 \text{ m}, \quad 112 \rightarrow 20 \text{ m}.$$

3 Multilooking, and why it must be done in linear power

Radar backscatter is multiplicative and is delivered as a logarithm. To average N pixels correctly you must average the *power*, then take the log:

$$P(p, u, v) = 10^{\sigma_{dB}(p, u, v)/10}$$

$$T_\ell(p, k) = 10 \log_{10} \left(\frac{1}{|B_k|} \sum_{(u, v) \in B_k} P(p, u, v) \right), \quad |B_k| = f_\ell^2.$$

Averaging in dB instead computes $\overline{10 \log_{10} P}$, the **geometric** mean of power, which is a different quantity:

$$10 \log_{10}(\overline{P}) \neq \overline{10 \log_{10}(P)}.$$

Measured on AmeriFlux_CA-Cbo, the discrepancy between the two is a mean bias of **0.53 dB**, a maximum of **7.5 dB**, and $r = 0.952$. The error is largest in heterogeneous cells — exactly where the signal lives.

Each level must be computed from P independently. A coarse level cannot be obtained by pooling a finer T , for the same reason.

4 Grouping by relative orbit

Backscatter depends on the incidence angle, which is fixed per relative orbit. Mixing orbits injects viewing geometry into everything downstream. Sentinel-1's repeat cycle is exactly 12 days, so the relative orbit is constant modulo 12:

$$g(p) = \text{ord}(t_p) \bmod 12.$$

Two keys are **merged** only when their long-run means agree at the within-group reproducibility ceiling. That recovers Sentinel-1A and 1B flying the *same* relative orbit six days apart, while keeping genuinely different orbits separate. Measured on CA-Cbo, where the mod-12 key gives $\{0:116, 1:107, 7:108\}$ passes:

```

split-half r(c_k) WITHIN a group    (the ceiling)    0.9979  0.9981  0.9982

r(c_k) BETWEEN groups    0 vs 1    0.9632    different relative orbit -> keep apart
                        0 vs 7    0.9552    different relative orbit -> keep apart
                        1 vs 7    0.9952    AT the ceiling: S1A/S1B, same orbit -> MERGE

```

Groups with fewer than 20 passes are discarded (see §10).

5 The decomposition

Within one station, one orbit group and one polarisation, form the $n \times \ell^2$ matrix T_ℓ and define three means:

$$\mu = \frac{1}{n \ell^2} \sum_p \sum_k T_\ell(p, k), \quad r_p = \frac{1}{\ell^2} \sum_k T_\ell(p, k), \quad c_k = \frac{1}{n} \sum_p T_\ell(p, k),$$

and the remainder

$$d(p, k) = T_\ell(p, k) - r_p - c_k + \mu$$

Equivalently, a double-centring projection:

$$D = \left(I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\right) T_\ell \left(I_{\ell^2} - \frac{1}{\ell^2} \mathbf{1}\mathbf{1}^\top\right),$$

from which the defining properties are immediate:

$$\sum_k d(p, k) = 0 \quad \forall p, \quad \sum_p d(p, k) = 0 \quad \forall k.$$

The three components are orthogonal, so the variance splits exactly:

$$\text{Var}(T) = \underbrace{\text{Var}(r)}_{\text{season}} + \underbrace{\text{Var}(c)}_{\text{static map}} + \underbrace{\text{Var}(d)}_{\text{interaction}}.$$

This is a check, not an assumption. Verified numerically to 2×10^{-7} dB²:

```

AmeriFlux_CA-Cbo      orbit 0    n=116
Var(T) total          = 2.1386 dB^2
Var(r_p) season       = 0.4195    19.6 %
Var(c_k) static map   = 1.3618    63.7 %
Var(d) interaction    = 0.3573    16.7 %
sum of the three      = 2.1386    |sum - Var(T)| = 2.4e-07

ISMN_TxSON_CR1000-1   orbit 3    n=113    season 37.7 % static 50.1 % inter 12.2 %
ISMN_SCAN_Abrams      orbit 4    n=104    season 40.7 % static 21.7 % inter 37.7 %

```

6 What each component means physically

$$\sigma^0(p, k) = \underbrace{c_k}_{\text{what this place is}} + \underbrace{(r_p - \mu)}_{\text{how wet the whole tile is}} + \underbrace{d(p, k)}_{\text{how wet this spot is versus its own norm}}$$

- c_k — **permanent texture**. A wood is always bright, a smooth field always dark. Roughness and structure, not water. It never changes.
- $r_p - \mu$ — **catchment wetness state** on date p . When it rains the entire tile brightens together.
- $d(p, k)$ — **differential wetting and drying**. What is left once season and permanent texture are removed.

A worked example on real CA-Cbo tokens (orbit 0, VV), three cells and two dates:

```

STEP 0  raw token sigma0 (dB)          tile mean r_p
20171217  -17.42  -9.70  -6.24  -10.05
20200710   -6.85  -6.52  -3.47   -6.88
grand mean mu = -8.40 dB
per-location mean c_k = -12.97  -8.15  -4.29

STEP 1  subtract the TILE MEAN FOR THAT DATE

```

20171217	-7.37	0.35	3.81
20200710	0.02	0.36	3.41
STEP 2 subtract the PER-LOCATION MEAN, add back mu -> this is d			
20171217	-2.80	0.10	-0.30
20200710	4.60	0.11	-0.70

The left-hand cell swings 10.6 dB in the raw data, but it is a permanently dark cell ($c_k = -12.97$), so most of that is simply what it is. After both subtractions it goes $-2.80 \rightarrow +4.60$: drier than its own norm on one date, much wetter on the other. The centre cell meanwhile barely moves, $+0.10 \rightarrow +0.11$ — it just tracks the tile.

Note there is no separate “seasonal mean” to compute. r_p , the tile-wide mean *on that date*, **is** the season. The two subtractions commute.

7 Why the decomposition rather than raw σ^0

The three components sum back to raw σ^0 up to a constant, so feeding all three is **lossless**: everything raw contains is present, merely factored.

The reason to factor it is that c_k **cannot be derived from a single sample**. It requires the multi-year archive. A network given raw σ^0 could only recover c_k by memorising a per-pixel mean for each of 993 stations — which is precisely the memorisation the design is trying to avoid, and which does not transfer to a station the model has never seen.

Computing c_k offline injects information the network cannot otherwise obtain, and it *does* transfer, because at inference c_k is computed from the new station’s own archive.

Worth testing raw σ^0 as an ablation, specifically on out-of-sample stations. The prediction is that raw matches on train and collapses out of sample, because that is where a memorised c_k stops working.

8 Normalisation

Each quantity is standardised within its station-orbit group, so that no station dominates:

$$\tilde{d} = \frac{d}{\text{sd}(d)}, \quad \tilde{c} = \frac{c - \bar{c}}{\text{sd}(c)}, \quad w_p = \frac{r_p - \bar{r}}{\text{sd}(r)}.$$

w_p is the catchment wetness state as a single standardised scalar per date.

This is unrelated to the global z-score TerraMind applies at `precompute_terr mind.py:140`, which uses one fixed constant (S1RTC mean $[-10.93, -17.329]$, sd $[4.391, 4.459]$) for every pixel of every image. A global affine map cannot remove a term that varies by date, nor one that varies by location. The two operations do not overlap.

9 Missing cells

With a validity mask $m(p, k) \in \{0, 1\}$ the single-pass centring is no longer exact. Alternate masked removals until both margins vanish:

$$d^{(i+\frac{1}{2})} = d^{(i)} - \frac{\sum_k m d^{(i)}}{\sum_k m}, \quad d^{(i+1)} = d^{(i+\frac{1}{2})} - \frac{\sum_p m d^{(i+\frac{1}{2})}}{\sum_p m}.$$

Terminate when $\max_p |\bar{d}(p, \cdot)| < 10^{-4}$ and $\max_k |\bar{d}(\cdot, k)| < 10^{-4}$. Typically 3–5 iterations. Measured missing-cell rates are small:

station	N	cell invalid%	dates w/ any bad	orbits (n>=20)
AmeriFlux_CA-Cbo	332	0.000%	0.0%	[0, 1, 7]
ISMN_TxSON_CR1000-1	114	0.000%	0.0%	[3]
ISMN_SCAN_Abrams	275	0.000%	0.0%	[4, 10]
ISMN_SCAN_AdamsRanch#1	239	0.201%	0.4%	[0, 6]

10 How many passes c_k needs

c_k is a mean over n passes, so it carries sampling error. Measured on CA-Cbo orbit 0, where the spread of c_k across cells is 1.167 dB:

passes used	$r(c_{\text{sub}}, c_{\text{full}})$	RMSE (dB)	as % of the c_k spread
5	0.9791	0.351	30.1
10	0.9897	0.257	22.0
20	0.9954	0.169	14.5
30	0.9972	0.137	11.7
50	0.9988	0.089	7.6
80	0.9996	0.051	4.3

$n \geq 50$ is comfortable; $n \geq 20$ is the floor. Medians are 212 (ascending) and 238 (descending), so almost every station qualifies. The degrees of freedom consumed by fitting r_p and c_k are 5.5 % at $n = 20$ and 1.5 % at $n = 100$ — negligible.

c_k should also be checked for seasonal balance. Pass counts are 22–28 % per season at every station examined, so the column mean is a genuine climatology rather than a summer-weighted one. A station with a lopsided record would give a c_k that absorbs part of the interaction signal.

11 Note on grid alignment

Within one tile, all cells share the same acquisition dates. One Sentinel-1 acquisition is a single 224×224 scene covering the whole tile: data is (N, 2, 224, 224) and dates is (N,) — one entry per *image*, not per location. T_ℓ is therefore a complete $n \times \ell^2$ matrix and c_k is a plain column mean over it. No temporal alignment step exists or is needed.

Dates do differ across stations, and across orbit groups within a station. Both are harmless, because centring is always performed *within* one station and one orbit group; c_k is never compared across stations, and $\tilde{d}$ is standardised per group.

12 What is handed to each decoder stage

Per sample, using the most recent available pass p^* :

stage	grid	cell	channels
bottleneck	14×14	160 m	$\tilde{c}_{14}, \tilde{d}_{14}(p^*), w_{p^*}$
up1	28×28	80 m	$\tilde{c}_{28}, \tilde{d}_{28}(p^*), w_{p^*}$
up2	56×56	40 m	$\tilde{c}_{56}, \tilde{d}_{56}(p^*), w_{p^*}$
up3	112×112	20 m	$\tilde{c}_{112}, \tilde{d}_{112}(p^*), w_{p^*}$

w_{p^*} is a scalar broadcast across the grid. Two housekeeping scalars accompany them: the age of p^* in days, and a validity flag which is zero when no usable pass exists — in which case $\tilde{d} = 0$, the natural “behaving like the tile average” value.

Concatenation alongside the existing skip, at `model.py:257-264`:

```

x = self.bottle_proj(bottleneck)                # (B, 512, 14, 14)
x = torch.cat([x, c_14, d_14, w], 1)             # 512 + 3

x = self.up1(x)                                  # 14 -> 28
x = self.conv1(torch.cat([x, film9(skip_L9), c_28, d_28, w], 1)) # 512+512+3 -> 256
x = self.up2(x)                                  # 28 -> 56
x = self.conv2(torch.cat([x, film6(skip_L6), c_56, d_56, w], 1)) # 256+256+3 -> 128
x = self.up3(x)                                  # 56 -> 112
x = self.conv3(torch.cat([x, film3(skip_L3), c_112, d_112, w], 1)) # 128+128+3 -> 64

```

Each `in_channels` grows by 3; `_ConvBlock` (`model.py:171`) is otherwise untouched. Added parameters are $3 \times C_{\text{out}} \times 9$ per stage: $6912 + 3456 + 1728 \approx 12\text{ k}$.

The weights on the new channels are zero-initialised,

```
for c in (conv1, conv2, conv3):
    c.block[0].weight[:, -3:] = 0
```

so at initialisation they contribute exactly nothing, the forward pass is bit-identical to `best.pt`, and \$26's provenance gate applies unchanged. Training moves them off zero.

13 Storage and build cost

Per station $\times$ orbit group $\times$ polarisation, with $n \approx 200$ passes:

$$(14^2 + 28^2 + 56^2 + 112^2) \times n = 16,660 \times 200 \approx 3.3 \times 10^6 \text{ values,}$$

which at fp16 is 6.6 MB, so roughly **13 GB** across 993 stations. The c_ℓ arrays are one per level and negligible.

Build cost is dominated by reading the rasters: $993 \times 2 \times \sim 66\text{ MB} \approx 130\text{ GB}$. One sbatch with `Pool(64)` and `--cpus-per-task=64`, approximately 30–60 minutes.

14 Summary

```
RAW          sigma0(p,k)  in dB, (N,2,224,224) per station per orbit direction
|
| P = 10^(sigma/10)          to LINEAR power
| mean over each f x f block, back to dB  MULTILOOK to level l
v
T_l(p,k)      (n, l^2) in dB
|
| group by ord(t_p) mod 12      RELATIVE ORBIT
| merge keys at the split-half ceiling
| drop groups with n < 20
v
per group: mu, r_p, c_k
|
| d = T - r_p - c_k + mu      DOUBLE-CENTRE
| iterate if cells are missing
| standardise
v
FED TO STAGE l:  c~_l  (permanent texture,      per cell)
                  d~_l  (today's anomaly,      per cell)
                  w_p   (catchment wetness state, scalar)
```